

[1] Index

Rev	Date	Designer	Description
A	25.01.26	Ganghyeok Lim	Create design project

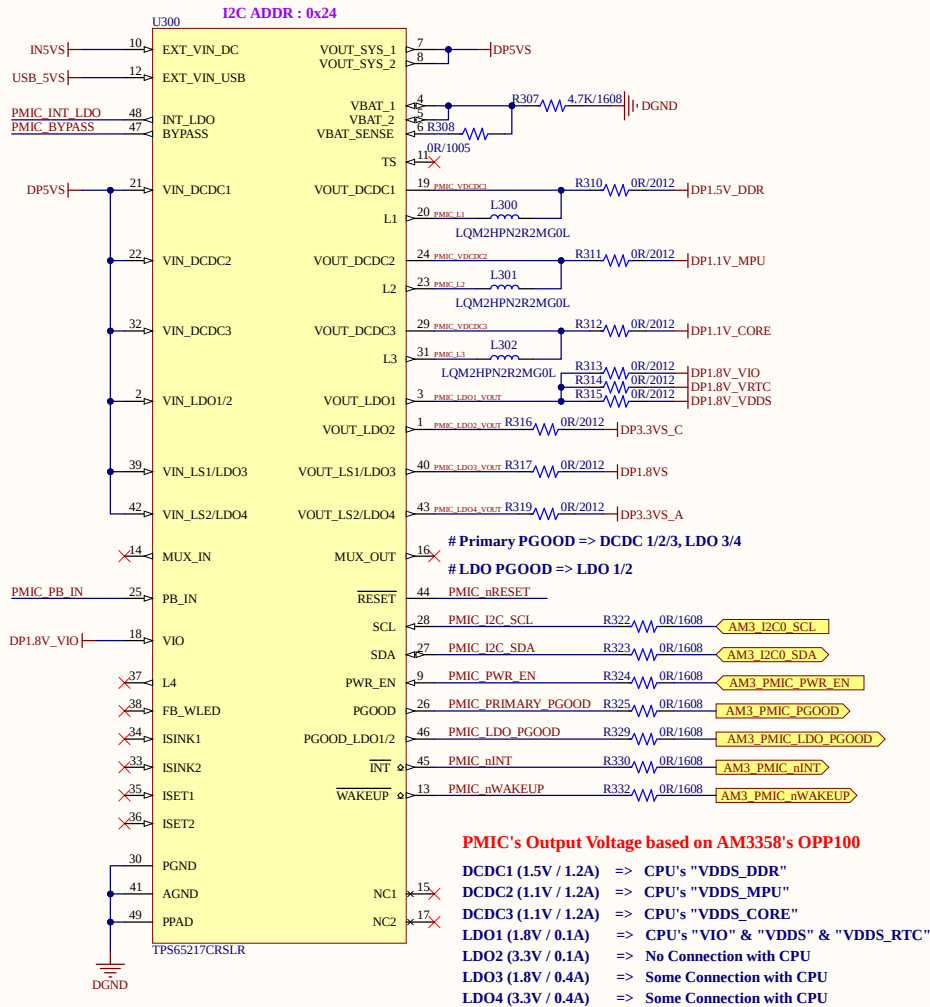
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#2 Overview	#9 ??
#3 ??	
#4 ??	
#5 ??	
#6 ??	
#7 ??	

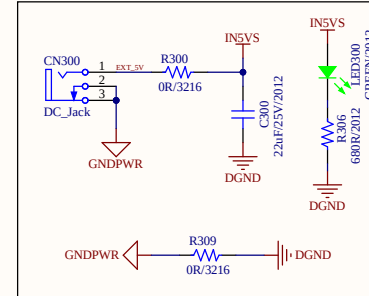
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Sheet # 1 of 11	Engineer Ganghyeok Lim	
Date 2025-01-26		
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A	Sitara Project B'd (Rev.A) [2] Overview																															
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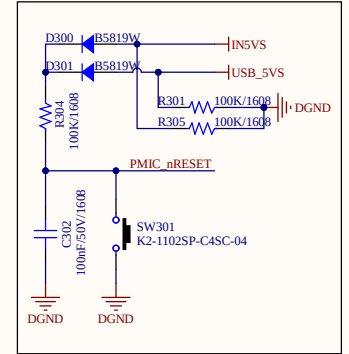
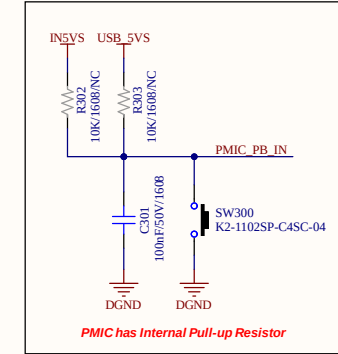
[3] Power



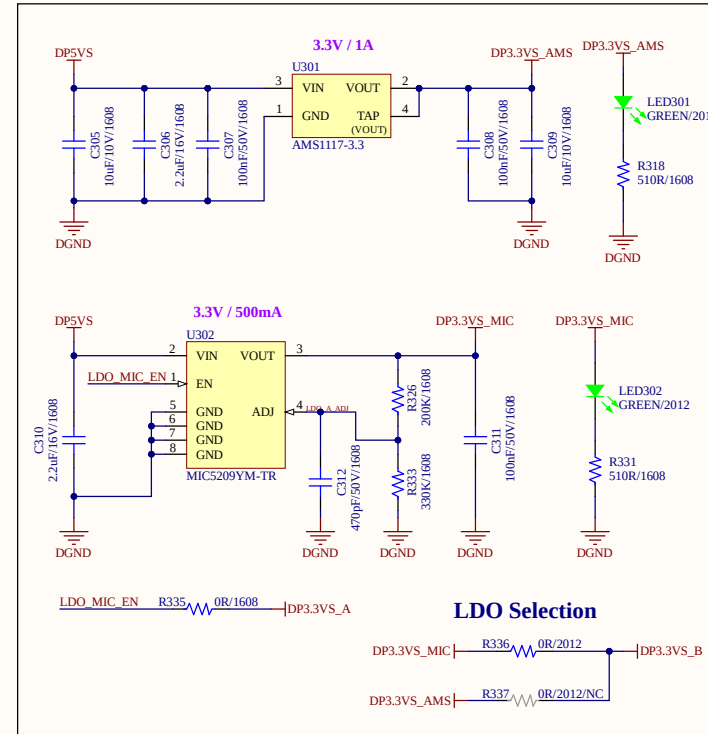
Power Connector (5V)



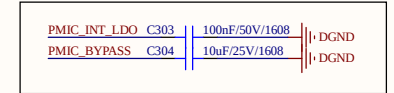
PMIC Power-On Button PMIC Reset Button



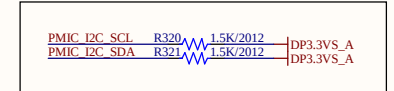
LDO (3.3V / 500mA)



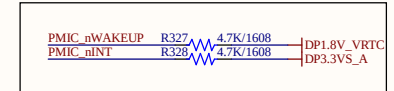
INT LDO / BYPASS Pin



I2C Pull-up Resistor



nWAKE-UP / nINT Pull-up

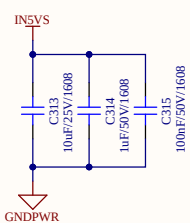


PMIC Power LED

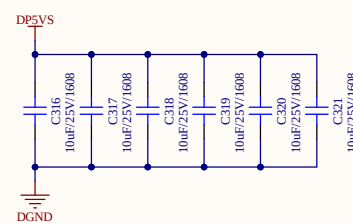


Decoupling Capacitors

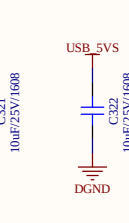
Decap for IN5VS



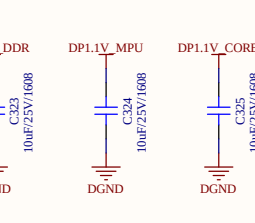
Decap for DP5VS



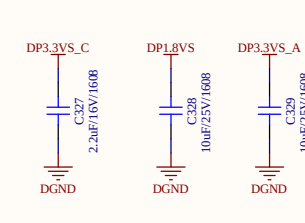
Decap for USB_5VS



Decap for DCDC 1/2/3

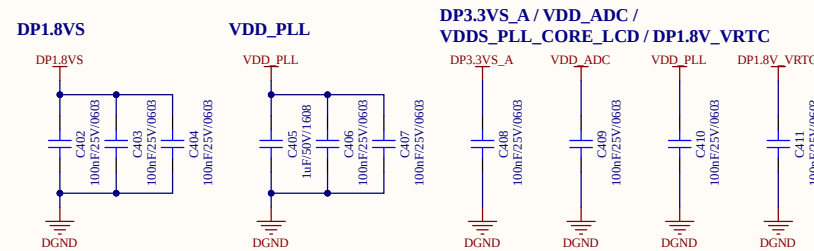
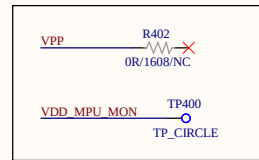
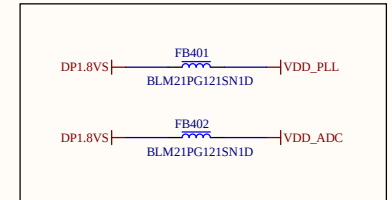
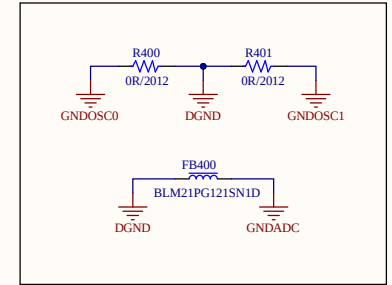
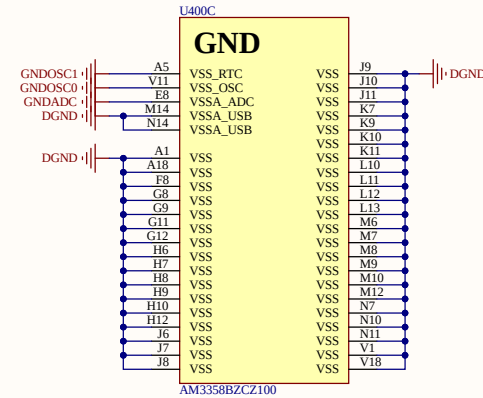
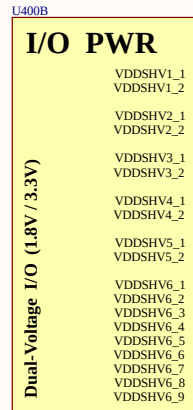


Decap for LDO 1~4



Place Decaps Closely to Each Pin of PMIC

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CAP_VDD_RTC	C400	1uF/50V/1608	DGND
CAP_VBB_MPU	C401	1uF/50V/1608	DGND
CAP_VDD_SRAM_MPU	C412	1uF/50V/1608	DGND
CAP_VDD_SRAM_CORE	C413	1uF/50V/1608	DGND

DP1.1V_MPU

DP1.1V MPU

C414 100nF/25V/0603

C415 100nF/25V/0603

C416 100nF/25V/0603

C417 100nF/25V/0603

C418 100nF/25V/0603

C419 100nF/25V/0603

DGND

DP1.8V_VDD5

DP1.8V VDD5

C420 100nF/25V/0603

C421 100nF/25V/0603

C422 100nF/25V/0603

C423 100nF/25V/0603

C424 100nF/25V/0603

C425 100nF/25V/0603

C426 100nF/25V/0603

DGND

DP3.3VS_A

DP3.3VS_A

C427 100nF/25V/0603

C428 100nF/25V/0603

C429 100nF/25V/0603

C430 100nF/25V/0603

C431 100nF/25V/0603

C432 100nF/25V/0603

C433 100nF/25V/0603

C434 100nF/25V/0603

C435 100nF/25V/0603

C436 100nF/25V/0603

C437 100nF/25V/0603

C438 100nF/25V/0603

C439 100nF/25V/0603

C440 100nF/25V/0603

C441 100nF/25V/0603

C442 100nF/25V/0603

C443 100nF/25V/0603

C444 100nF/25V/0603

C445 100nF/25V/0603

C446 100nF/25V/0603

C447 100nF/25V/0603

DGND

DP1.1V_CORE

DP1.1V CORE

C448 100nF/25V/0603

C449 100nF/25V/0603

C450 100nF/25V/0603

C451 100nF/25V/0603

C452 100nF/25V/0603

C453 100nF/25V/0603

C454 100nF/25V/0603

C455 100nF/25V/0603

C456 100nF/25V/0603

C457 100nF/25V/0603

C458 100nF/25V/0603

C459 100nF/25V/0603

C460 100nF/25V/0603

C461 100nF/25V/0603

C462 100nF/25V/0603

C463 100nF/25V/0603

DGND

DP1.5V_DDR

DP1.5V DDR

C464 100nF/25V/0603

C465 100nF/25V/0603

C466 100nF/25V/0603

C467 100nF/25V/0603

C468 100nF/25V/0603

C469 100nF/25V/0603

C470 100nF/25V/0603

C471 100nF/25V/0603

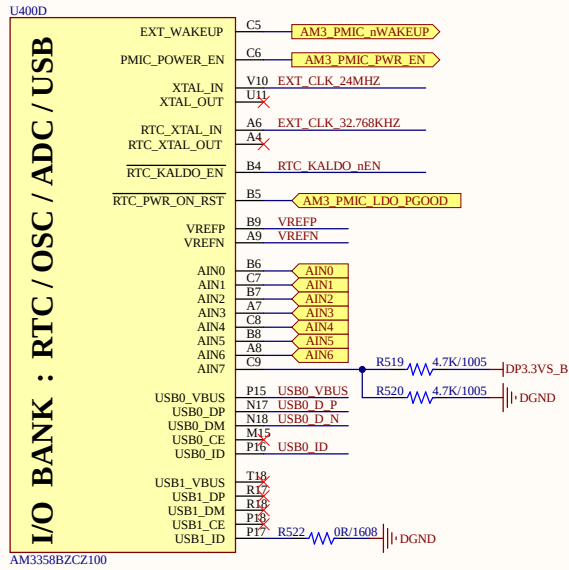
C472 100nF/25V/0603

DGND

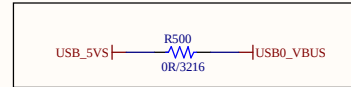
VDDSHV1~6

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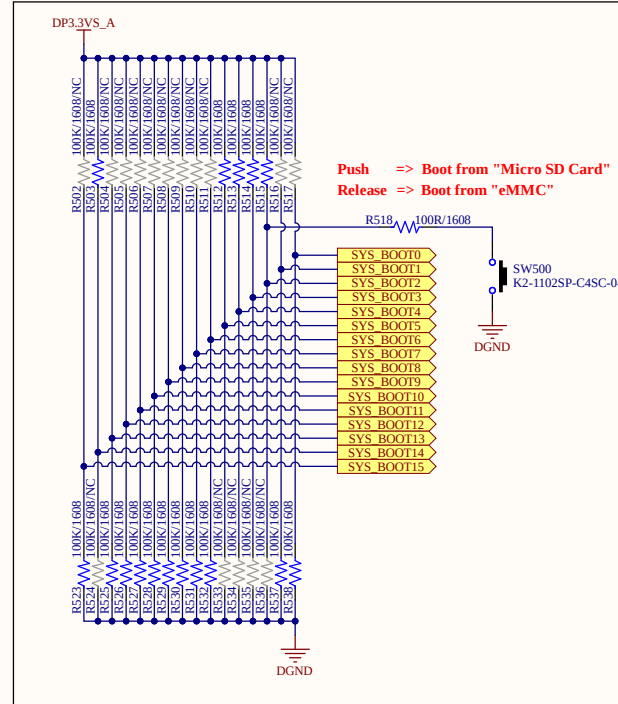
[5] Sitara AUX Part



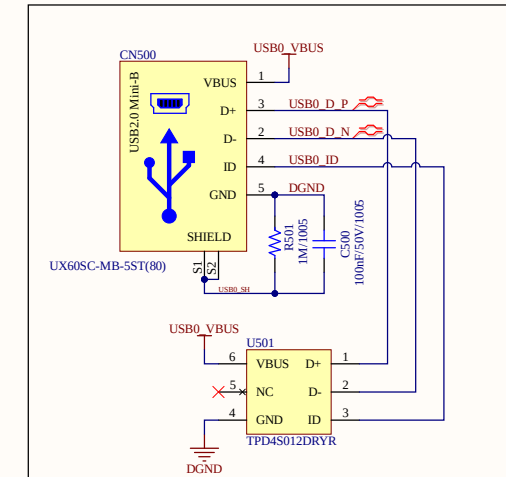
USB Power



Boot Config



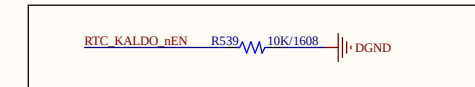
USB 2.0 Mini-B Connector



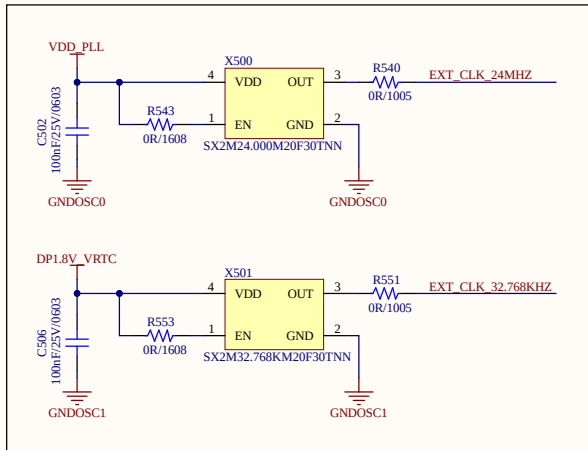
Indication LED (USB0 VBUS)



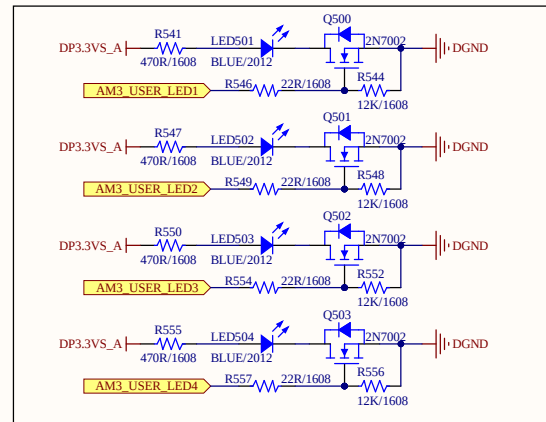
Internal RTC LDO Enable



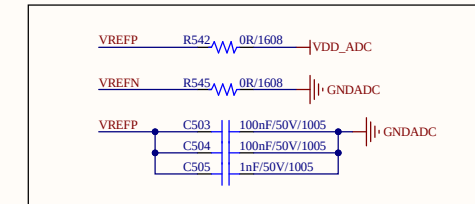
External Oscillator (24MHz / 32.768kHz)



User LEDs



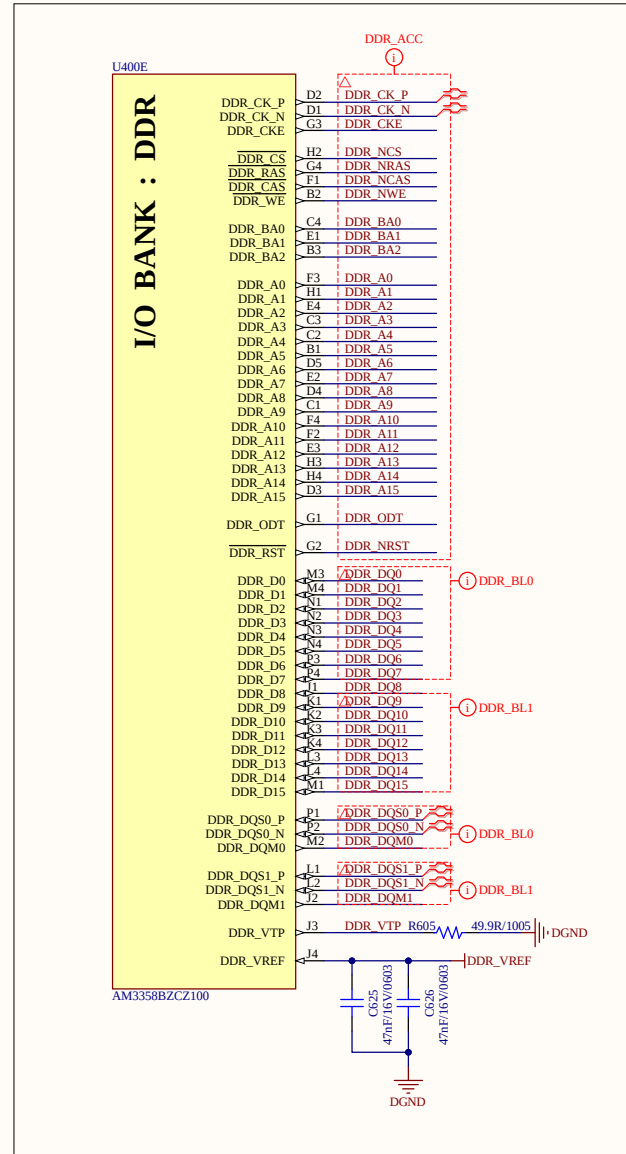
Analog Reference



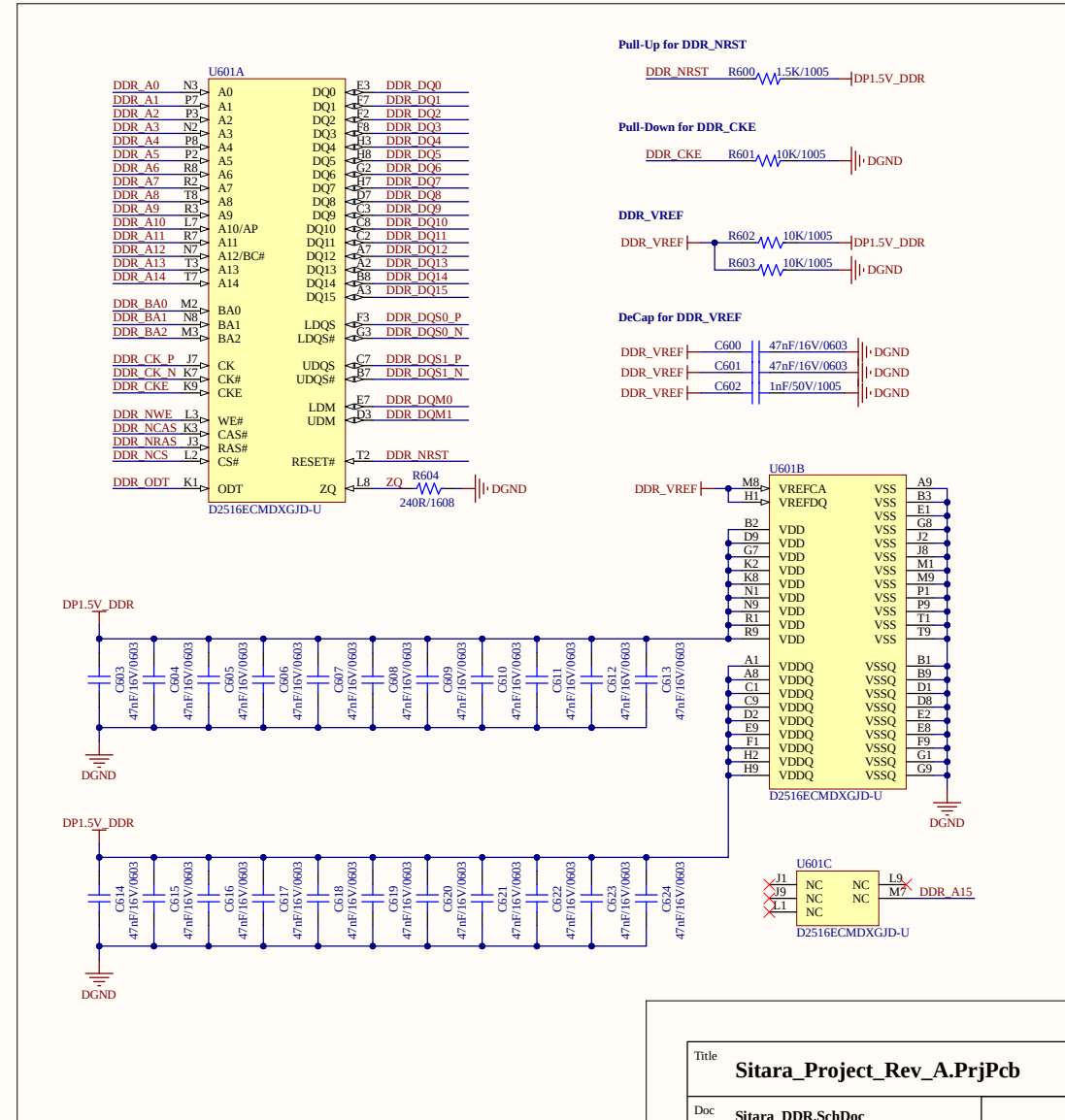
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[6] Sitara DDR Interface

Sitara DDR Memory Controller Interface (EMIF)



DDR3L Memory



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The diagram illustrates the hardware connection for the AT24C16 I2C EEPROM. The AT24C16 is a 16-bit serial EEPROM with pins for VCC, GND, SDA, SCL, and WP. It is connected to the AM3's I2C0_SDA and I2C0_SCL pins. The VCC pin is connected to the DP3.3VS_A supply through a 4.7k resistor and a 100nF capacitor. The WP pin is connected to the DP3.3VS_A supply through a 4.7k resistor. The SDA and SCL pins are connected to the AM3's I2C0_SDA and I2C0_SCL pins respectively. The diagram also shows the AT24C16's internal structure and the AM3's I2C0_SDA and I2C0_SCL pins.

Place Serial Resistors Close to MPU

DP3.3VS_B

MMC0_CLK R710 22R/1005

MMC0_CMD R712 22R/1005

MMC0_DAT0 R713 22R/1005

MMC0_DAT1 R714 22R/1005

MMC0_DAT2 R715 22R/1005

MMC0_DAT3 R716 22R/1005

DP3.3VS_B

Micro SD CN700

VDD 4

CLK 5

CD 6

DAT0 7

DAT1 8

DAT2 9

DAT3/CD 10

VSS 11

TF-01A

DGND

DP3.3VS_B

DP3.3VS_B

C701 10uF/25V/1608

C702 100nF/16V/1005

DGND

DGND

The schematic diagram illustrates the electrical connections for the MTF32GAZAQHD-WT board, featuring two main integrated circuits: U703A and U702.

U703A (MTFC32GAZAQHD-WT) Pin Connections:

- VCC1, VCC2, VCC3, VCC4:** Connected to DP3_3VS_B.
- VCCQ1, VCCQ2, VCCQ3, VCCQ4, VCCQ5:** Connected to DP1.8VS.
- VSS1, VSS2, VSS3, VSS4, VSS5, VSS6:** Connected to DGND.
- VSSQ1, VSSQ2, VSSQ3, VSSQ4, VSSQ5:** Connected to DGND.
- CLK, RST, CMD:** Connected to M6, K5, and M5 respectively.
- DAT0, DAT1, DAT2, DAT3, DAT4, DAT5, DAT6, DAT7:** Connected to A3, A4, A5, B2, B3, B4, B5, and B6 respectively.
- DS:** Connected to H5.
- VDDIM:** Connected to C2.

U702 (SN74LVC1G06DCKR) Pin Connections:

- VCC:** Connected to DP1.8VS.
- GND:** Connected to DGND.
- NC:** Connected to DP1.8VS.
- Y:** Connected to MMC1_nRST.
- MMC1_RST:** Connected to GND.
- MMC1_nRST:** Connected to MMC1_CMD.

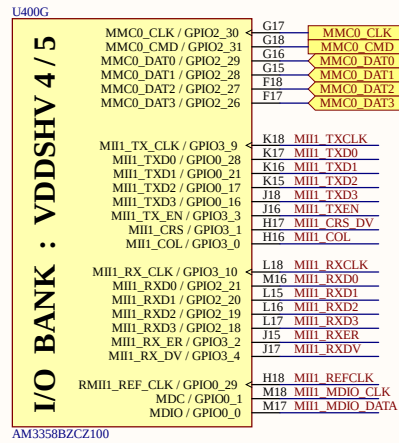
Passive Components:

- Resistors:** R719, R720, R721, R718, R723, R732, R741, R733, R734, R735, R736, R737, R738, R739, R740, R742, R743, R744, R745, R746, R747, R748, R749, R750, R751, R752, R753, R754, R755, R756, R757, R758, R759, R760, R761, R762, R763, R764, R765, R766, R767, R768, R769, R770, R771, R772, R773, R774, R775, R776, R777, R778, R779, R780, R781, R782, R783, R784, R785, R786, R787, R788, R789, R790, R791, R792, R793, R794, R795, R796, R797, R798, R799, R800, R801, R802, R803, R804, R805, R806, R807, R808, R809, R810, R811, R812, R813, R814, R815, R816, R817, R818, R819, R820, R821, R822, R823, R824, R825, R826, R827, R828, R829, R830, R831, R832, R833, R834, R835, R836, R837, R838, R839, R840, R841, R842, R843, R844, R845, R846, R847, R848, R849, R850, R851, R852, R853, R854, R855, R856, R857, R858, R859, R860, R861, R862, R863, R864, R865, R866, R867, R868, R869, R870, R871, R872, R873, R874, R875, R876, R877, R878, R879, R880, R881, R882, R883, R884, R885, R886, R887, R888, R889, R890, R891, R892, R893, R894, R895, R896, R897, R898, R899, R900, R901, R902, R903, R904, R905, R906, R907, R908, R909, R910, R911, R912, R913, R914, R915, R916, R917, R918, R919, R920, R921, R922, R923, R924, R925, R926, R927, R928, R929, R930, R931, R932, R933, R934, R935, R936, R937, R938, R939, R940, R941, R942, R943, R944, R945, R946, R947, R948, R949, R950, R951, R952, R953, R954, R955, R956, R957, R958, R959, R960, R961, R962, R963, R964, R965, R966, R967, R968, R969, R970, R971, R972, R973, R974, R975, R976, R977, R978, R979, R980, R981, R982, R983, R984, R985, R986, R987, R988, R989, R990, R991, R992, R993, R994, R995, R996, R997, R998, R999, R1000.
- Capacitors:** C709, C704, C705, C706, C707, C708, C710, C711, C712, C713, C714.

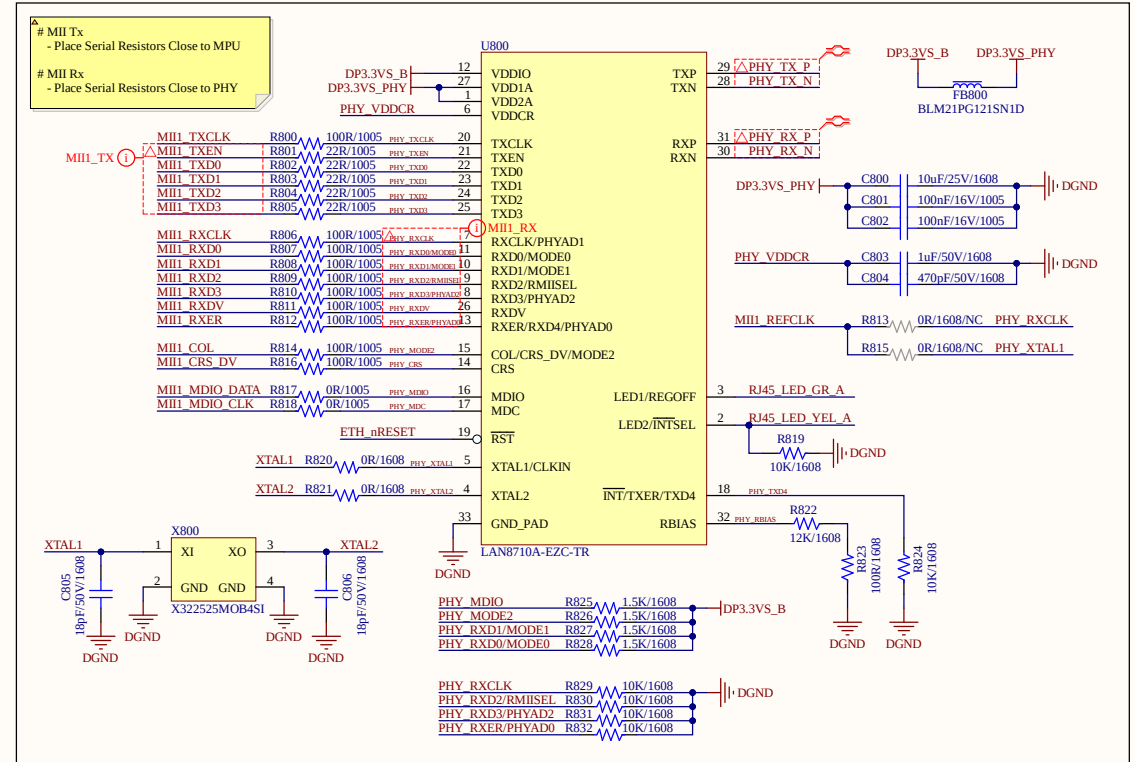
Table of Components:

U703A	U702	U703B	U703C	U703D	U703E	U703F	U703G	U703H	U703I	U703J	U703K	U703L	U703M	U703N	U703O	U703P	U703Q	U703R	U703S	U703T	U703U	U703V	U703W	U703X	U703Y	U703Z
NC61	NC62	NC63	NC64	NC65	NC66	NC67	NC68	NC69	NC70	NC71	NC72	NC73	NC74	NC75	NC76	NC77	NC78	NC79	NC80	NC81	NC82	NC83	NC84	NC85	NC86	NC87

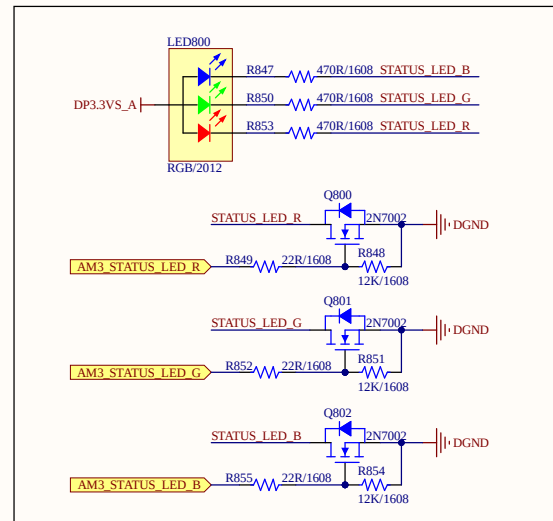
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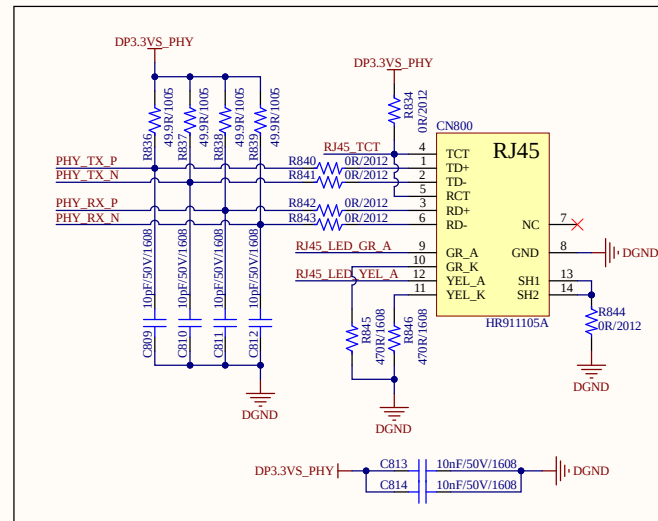
Ethernet PHY (10/100Mbps)



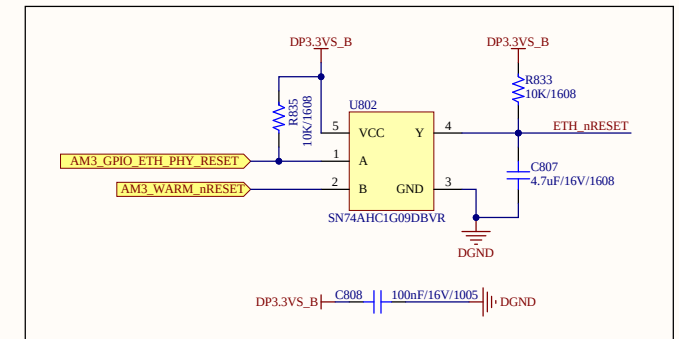
User LEDs



RJ45 Connector

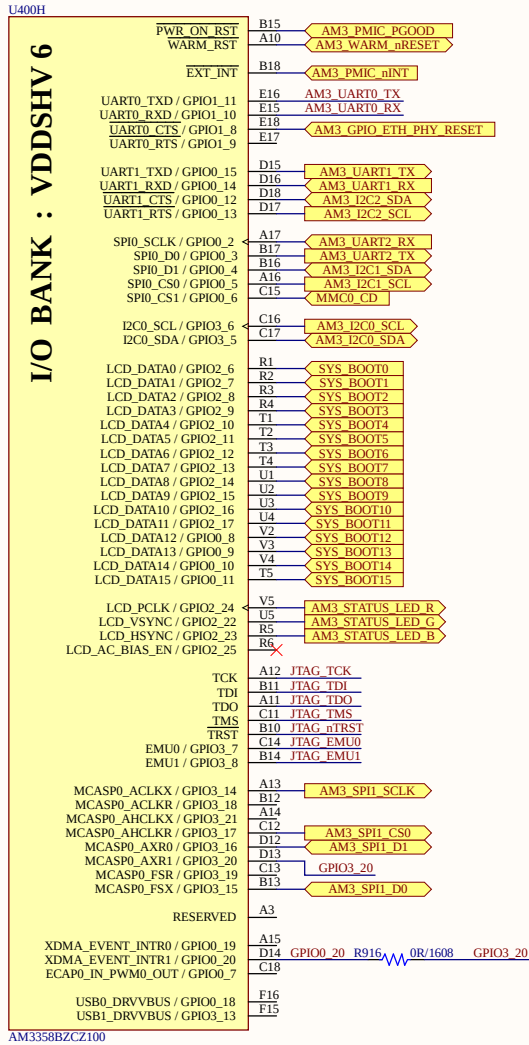


Ethernet PHY Reset

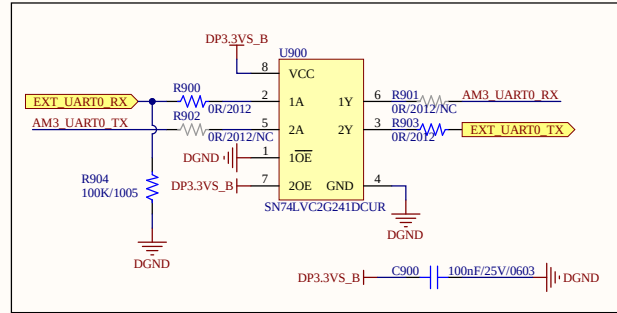


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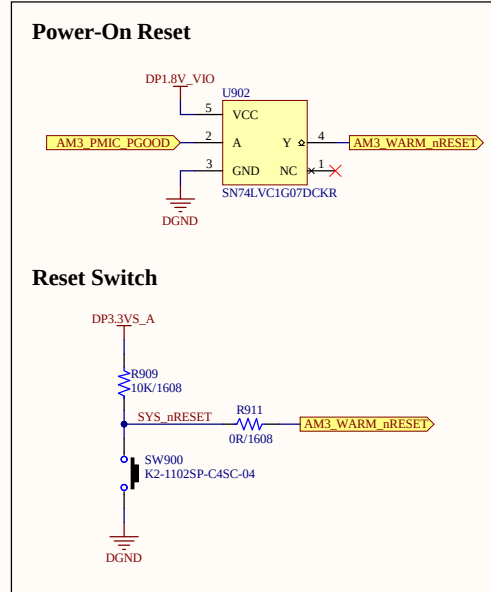
[9] Sitara I/O Bank 6



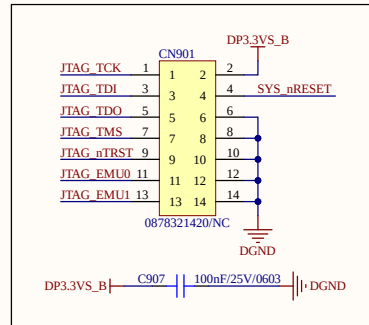
UART0 External Connector



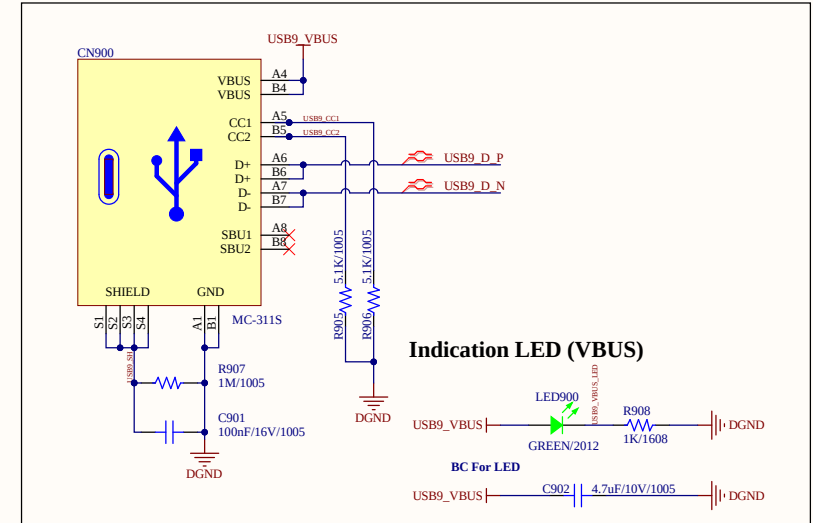
System Reset



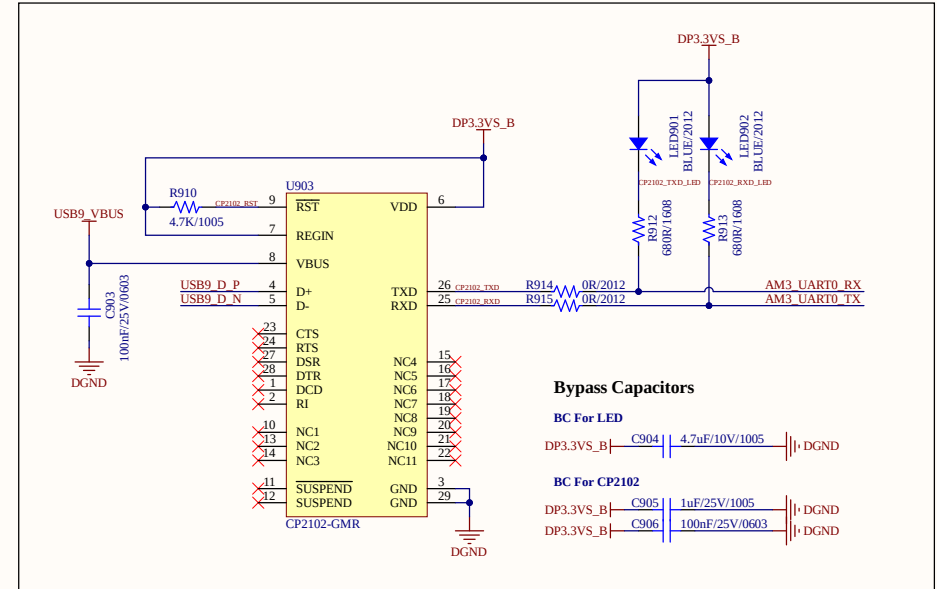
JTAG Connector



USB Connector (Type-C)

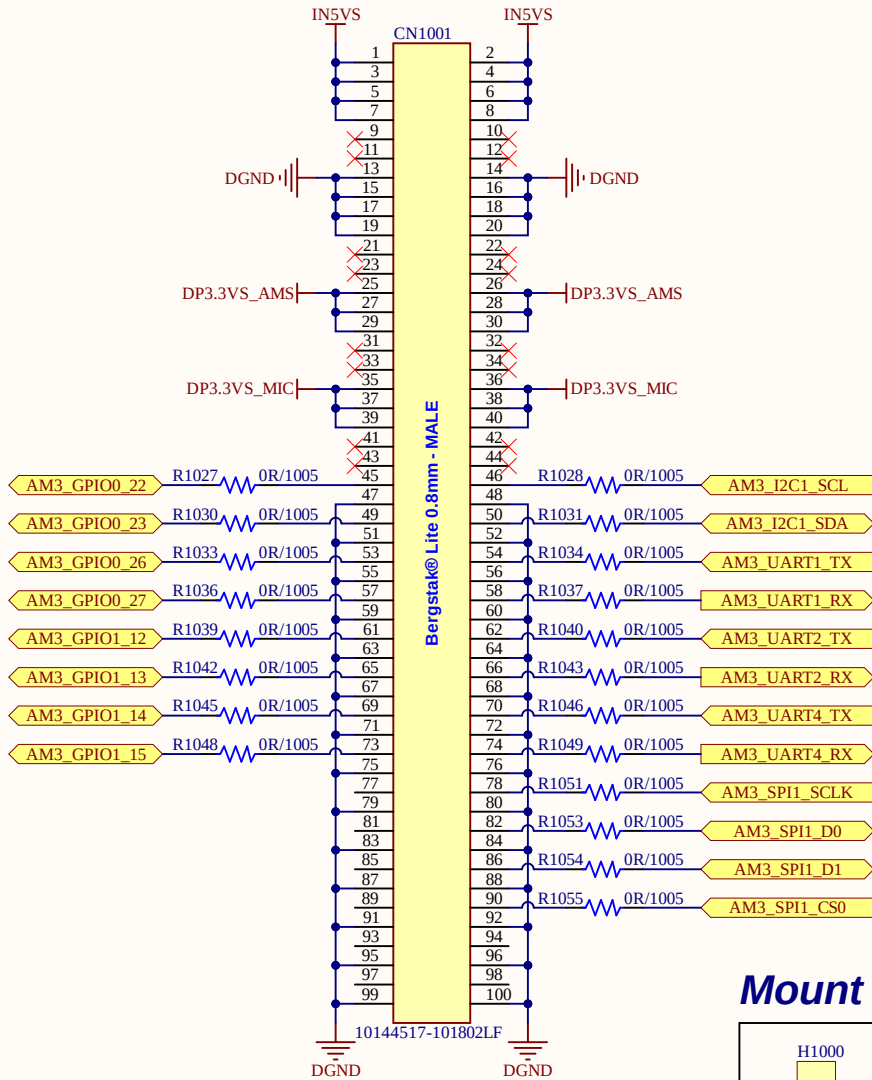


USB - UART Bridge

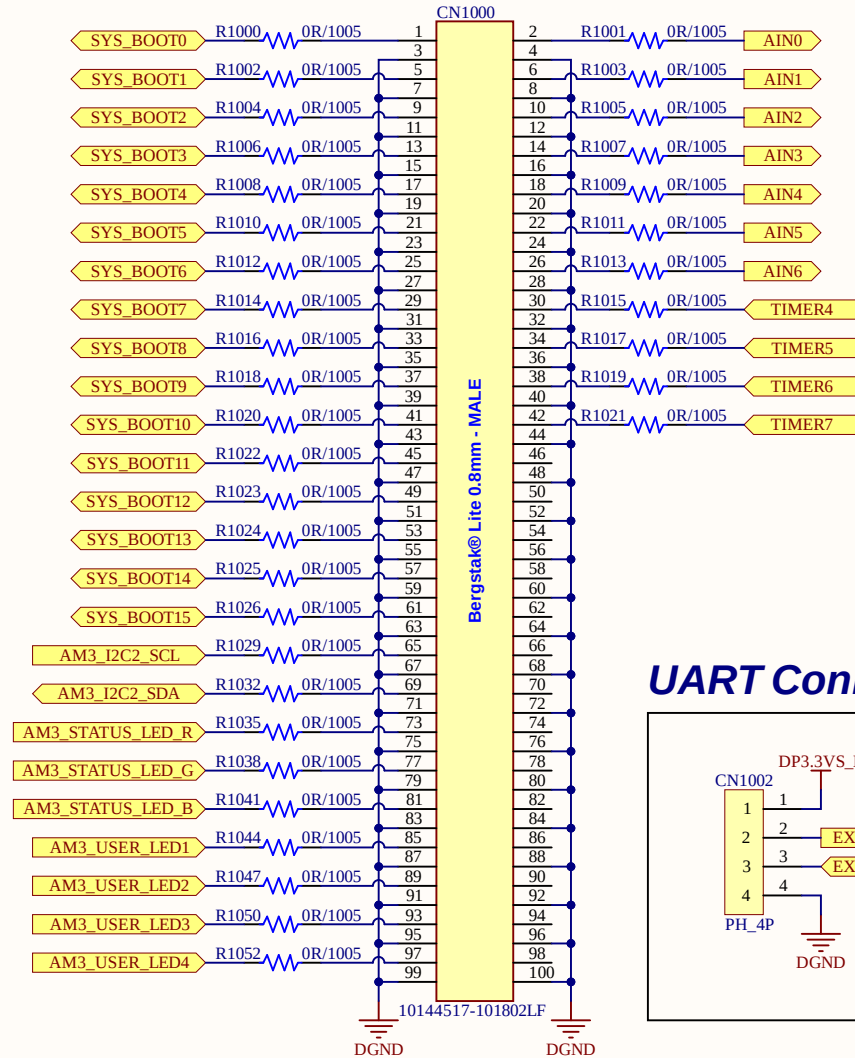
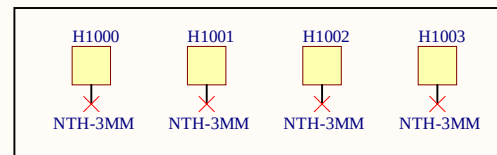


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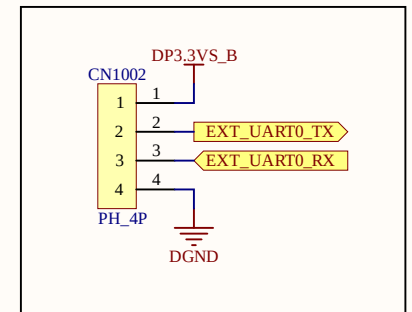
[10] Connectors



Mount Holes



UART Connector



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Sitara_Connectors.SchDoc		
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[?] Schematic Design Check-List

Schematic Design Check-List

No.	Category	Item	Description	Confirm

